



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	Earlier today, the company's systems were subjected to a Denial-of-Service (DoS) attack, causing an internal network outage that lasted for two hours. During the attack, a massive influx of ICMP packets overwhelmed the systems, causing network services to become unresponsive and preventing legitimate internal traffic from accessing network resources. A subsequent investigation by the cybersecurity team revealed that attackers had exploited a misconfigured firewall to flood the corporate network with ICMP ping requests, resulting in a system crash. To address this incident, the cybersecurity team implemented firewall rules to limit ICMP request rates and validate source IPs, deployed an Intrusion Detection System (IDS) to filter suspicious traffic, and installed network monitoring software to detect anomalous traffic patterns. The company experienced a security event when all network services suddenly stopped responding. The cybersecurity team found the disruption was caused by a distributed denial of services (DDoS) attack through a flood of incoming ICMP packets. The team responded by blocking the attack and stopping all non-critical network services, so that critical network services could be restored. Comment: summarize how the incident happen and how to respond to the
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	current situation.
Identify	The cybersecurity team audited the systems, devices, and access policies involved in the attack to identify security vulnerabilities. The team found a large volume of ICMP requests directed at company systems during the attack. Preliminary checks indicated that the systems were unable to function properly. A malicious actor or actors targeted the company with an ICMP flood attack. The entire internal network was affected. All critical network resources needed to be secured and restored to a functioning state.
Protect	To defend against future attacks, the team has implemented IDS/IPS systems and network monitoring software designed to detect anomalous traffic patterns. Furthermore, new firewall configurations have been deployed, including source IP filtering and rate-limiting for incoming ICMP packets.
Detect	To detect future unauthorized access attacks, the team will use firewall logging tools and intrusion detection systems (IDS) to monitor all incoming traffic from the internet.
Respond	The team has updated system security configurations, including the addition of IDS/IPS systems and enhanced firewall settings. We have informed senior management, who will send a letter to clients notifying them of the data breach. Additionally, in accordance with local laws and regulations, management will report the incident to law enforcement agencies and other relevant authorities. For future security events, the cybersecurity team will isolate affected systems to prevent further disruption to the network. They will attempt to restore any critical systems and services that were disrupted by the event. Then, the team will analyze network logs to check for suspicious and abnormal activity. The team will also report all incidents to upper management and appropriate legal authorities, if applicable.

	Comment: Addressing the current issue and preventing future occurrences
Recover	The team blocks incoming ICMP packets, stops all non-critical network services, and restores critical network services. To recover from a DDoS attack by ICMP flooding, access to network services need to be restored to a normal functioning state. In the future, external ICMP flood attacks can be blocked at the firewall. Then, all non-critical network services should be stopped to reduce internal network traffic. Next, critical network services should be restored first. Finally, once the flood of ICMP packets have timed out, all non-critical network systems and services can be brought back online. Comment: write it in more detail.

Reflections/Notes: Drawing lessons from this incident, we should conduct regular security reviews to promptly identify and patch system vulnerabilities. Furthermore, system security configurations should align with NIST standards, and a combination of security tools should be employed to enhance overall system security.